UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

END SEMESTER EXAMINATION
SEMESTER II - 2013/2014

General Physics II - PH 102

Time allowed: *Two hours*

Answer **FIVE** questions only selecting **TWO** from Section A, **TWO** from Section B and **ONE** from Section C.

(Use of calculators are allowed)

Electron charge (e)	$= 1.60 \times 10^{-19}$ C
Rest mass of electron (m_o)	$= 9.11 \times 10^{-31}$ kg
Permittivity of free space (ε_o)	$= 8.85 \times 10^{-12}$ F m^{-1}
Permeability of free space (μ_o)	$= 4\pi \times 10^{-7}$ H m^{-1}
Planck's constant (h)	$= 6.63 \times 10^{-34}$ J s
Speed of light (c)	$= 3.00 \times 10^{8}$ m s^{-1}
Boltzmann constant (k)	$= 1.38 \times 10^{-23}$ J K^{-1}
Avogadro number (N_A)	$= 6.02 \times 10^{23}$ mol^{-1}.

(Answer Section A in a separate book)

SECTION - A

1. (a) Clearly identifying the quantities involved, write down the Gauss' law.

(b) An infinitely long straight wire of radius R has a uniform linear charge density λ (C m^{-1}). Derive expressions for the electric field intensity at a distance r from the axis of the wire for the cases where $r > R$ and $r \leq R$.

(c) Sketch the variation of the electric field intensity with r.

(d) Suppose a long straight wire having a charge density of $\lambda = +5\times10^{-3}$ C m^{-1} is placed horizontally and a small mass of 0.2 mg having a charge of +54 μC is kept in equilibrium at a point vertically above the wire. Find the distance to the charge from the wire.

2. (a) Define self-inductance.
Obtain an expression for the self induced e.m.f. of a coil of self inductance L, which carries a time varying current I.

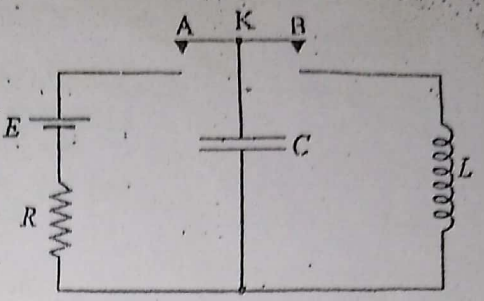
(b) Write down an expression for the energy stored in an inductor L, when a current I is passing through it.
A toroid having 100 turns has a mean inner radius of 5.0 cm and a circular cross section of area 1.0 cm^2.

(i) Determine the self inductance of the coil.

(ii) Find the energy stored in the toroid when a current of 2.0 A is passing through it.

CamScanner

3. A battery of e.m.f. E, a capacitor of capacitance C and a and a resistor of resistance R and an inductor of inductance L are connected to a two way key K as shown in the figure. Initially the capacitor has no charge and the end A of the key K is closed at time $t = 0$.



(a) Derive an expression for the charge q in the capacitor at any time t.

(b) Obtain an expression for the current i as a function of time t.

(c) Suppose $E = 6$ V, $R = 5$ kΩ and $C = 2.0$ mF.

(i) Calculate the time constant of the circuit and

(ii) Find charge of the capacitor and the current in the circuit at time $t = 0.1$ s.

(iii) How long is needed to charge the capacitor to its practical limit?

(d) After the capacitor is fully charged, the end B of the key K is suddenly pressed.

(i) Write down an expression for the resonant frequency of the circuit.

(ii) If the value of L is 8 mH calculate the resonant frequency of the circuit.

(ii) Draw the variation of current with time through the circuit in this part.

(Answer Sections B and C together in separate books)

SECTION B

4. (a) Briefly explain what is Compton shift.

(b) Derive the equation for the Compton shift $\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$. Identify the symbols used in the equation.

(c) Qualitatively sketch the graphs of λ' versus (1-cosθ) for X-rays and γ rays in the same diagram. Assume that the incident X-ray wavelength is λ_X and incident γ ray wavelength is λ_γ and $\lambda_X \neq \lambda_\gamma$.

(d) X-ray photons of wavelength 0.02480 nm are incident on a target and the Compton-scattered photons are observed at 90.0°.

(i) Calculate the wavelength of the scattered photons?

(ii) Calculate the momentum of the incident photons and the scattered photons?

(iii) Calculate the kinetic energy of the scattered electrons?

(iv) Calculate the momentum (magnitude and direction) of the scattered electrons?

CamScanner

5. (a)(i) State the Bohr's postulates.

(ii) Show that the quantized electron energy levels of a H-atom is given by $E_n = -\frac{me^4}{8\varepsilon_o^2 h^2}\left(\frac{1}{n^2}\right)$, where $n = 1, 2, 3, 4, \ldots\ldots$ and other symbols have their usual meanings.

(iii) Calculate the ionization energy of H-atom.

(iv) Modify the above equation for a H-like ion with Z protons in the nucleus.

(v) Find the shortest and longest wavelengths of the Lyman's series of a singly ionized He atom.

(vi) State two failures of the Bohr model of atoms.

(b)(i) Qualitatively sketch the X-ray spectrum of a given metal and explain how the continuous spectrum is produced.

(ii) Calculate the minimum X-ray wavelength produced in continuous (bremsstrahlung) spectrum by electrons that have been accelerated through 3.0×10^4 V.

6. (a) Write down three important features of the "strong force".

(b) List the conservation laws satisfied in nuclear reactions.

(c) Calculate the total binding energy, and the binding energy per nucleon, for $^{40}_{20}Ca$.

neutron mass = 1.007276 u, atomic mass of Ca = 40.078 u
proton mass = 1.008665 u Equivalent energy of 1 u is 931.5 MeV

(d) Consider the element X, $^{A}_{Z}X$, and briefly explain the three decay processes giving the basic decay equations.

(e) A physicist determines that a sample of dead wood has a carbon-14 decay rate of 6.00 counts per minute per gram. What is the age of the piece of wood in years? The decay rate of carbon-14 in fresh wood today is 13.6 counts per minute per gram, and the half life of carbon-14 is 5730 years.

CamScanner

SECTION – C

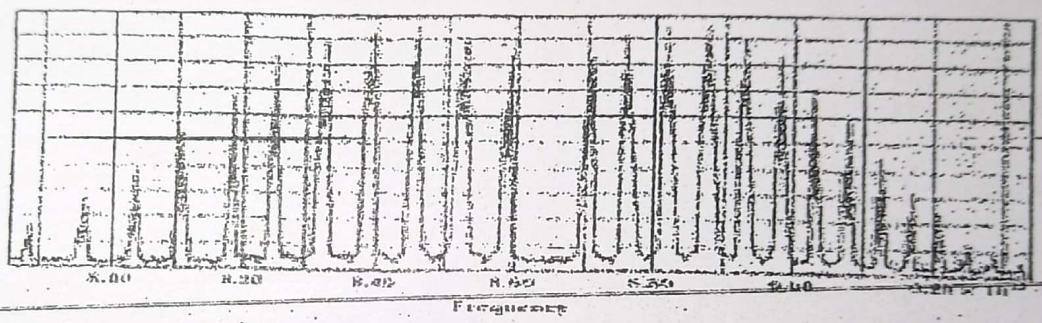
7. (a) Sketch qualitatively the potential energy vs interatomic distance curves for the following materials in the same diagram.

	Melting point (°C)	Thermal expansion coefficient (10^{-6} K^{-1})	Elastic modulus (10^9 N m^{-2})
Polymer	65	22	1
metal	660	108	70

(b) (i) Write down an equation for the rotational energy levels of a diatomic molecule (Identify the symbols you use).

(ii) Write down an equation for the vibrational energy levels of a diatomic molecule (Identify the symbols you use)

(iii) The absorption spectrum of HCl molecule in vapor phase is given below.



Considering both rotational and vibrational energies, identify the rotational and vibrational quantum numbers of the two levels involved with the missing transition (prohibited transition).

(iv) Calculate the energy associated with the missing transition. Assume that the molar masses of H and Cl are 1.01g mol^{-1} and 35.45 g mol^{-1} respectively. The force constant of HCl molecule is 480 N m^{-1}.

(v) In the above figure all absorption lines are split into two closely spaced lines. What is the reason for the splitting?

8. (i) Give labeled energy band diagrams for a metal, a semiconductor and an insulator and discuss the electrical behavior of these materials.

(ii) Sketch the variation of resistivity with temperature for a metal, a semiconductor and for a superconductor.

(iii) Use the classical free electron model to estimate the average speed of random thermal motion V_R of electrons in copper at room temperature (300 K). A copper wires of cross sectional area 1.0 mm^2, is connected to a circuit and the current flowing in the circuit is 1.0 A. Find the drift velocity V_d of free electrons in copper. Comment on the magnitudes of these motions (V_R and V_d) and state which component is responsible for the electrical conductivity. Assume that the density of free electron in copper is 10^{29} m^{-3}.

CamScanner